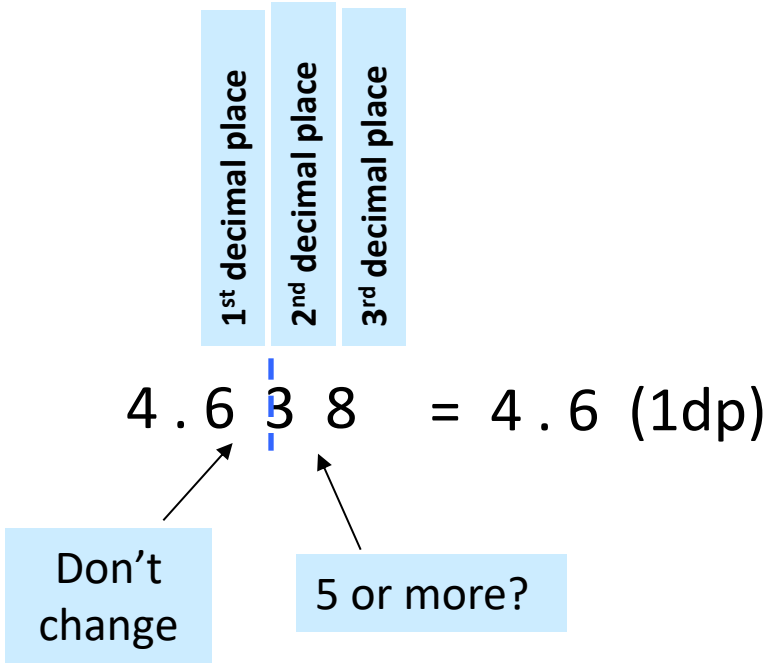
Rounding to decimal places

Eg round 4.638 to 1 decimal place



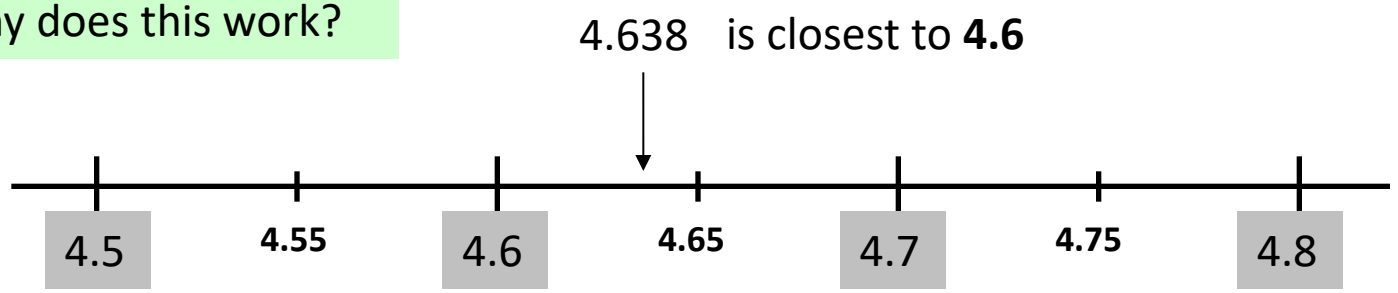
Imagine line after desired decimal place

If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

Write a rounded value with as many digits as there were before the line

Why does this work?



Rounding to decimal places

Eg round 23.4281 to 2 decimal places

1st decimal place | 2nd decimal place | 3rd decimal place

23 . 4 2 8 1 = 23.43 (2dp)

Increase by 1

5 or more?

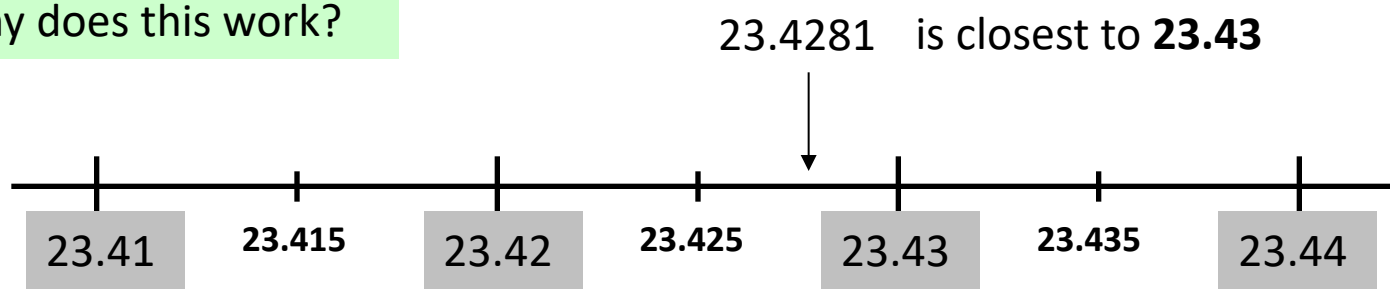
Imagine line after desired decimal place

If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

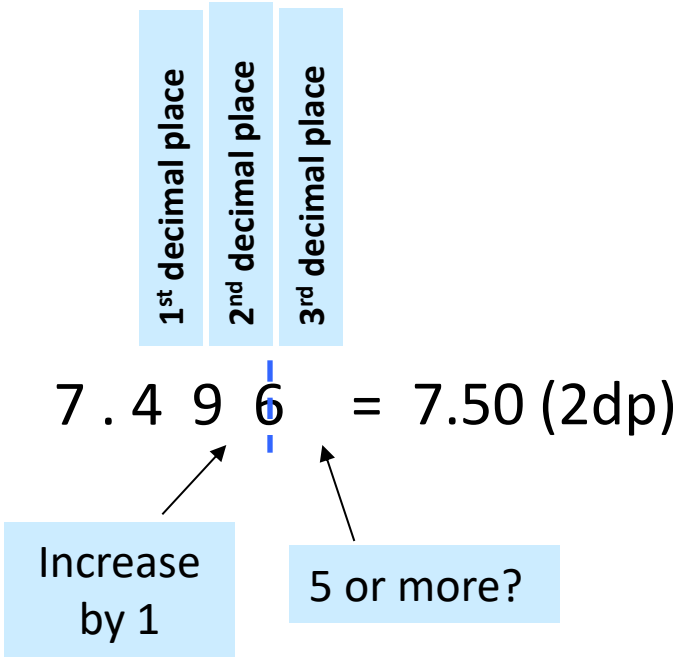
Write a rounded value with as many digits as there were before the line

Why does this work?



Rounding to decimal places

Eg round 7.496 to 2 decimal places



Imagine line after desired decimal place

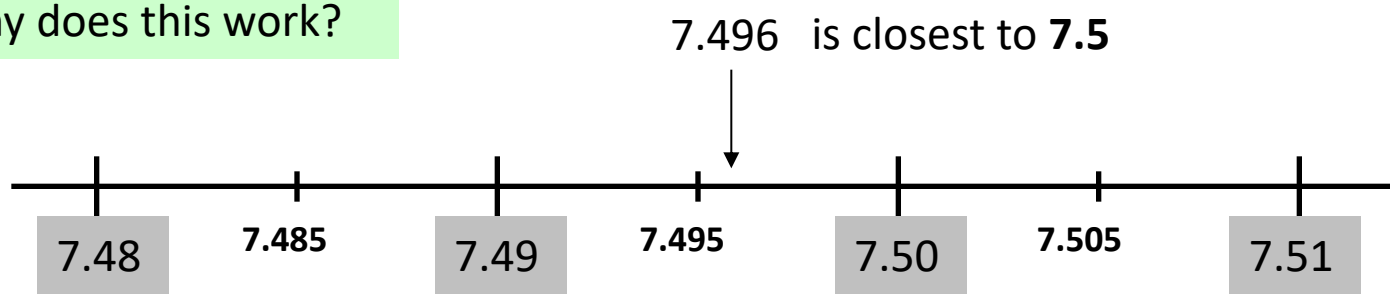
If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

If increasing the value makes it 10, you must carry into the next column, as you would with written addition

Write a rounded value with as many digits as there were before the line

Why does this work?





Rounding to decimal places



1. Round each number to the required number of decimal places (dp)

a) 7.842 (1dp) = _____

b) 45.65 (1dp) = _____

c) 0.109 (1dp) = _____

d) 4.817 (2dp) = _____

e) 12.969 (2dp) = _____

f) 0.0814 (2dp) = _____

g) 5.49284 (3dp) = _____

h) 0.007891 (3dp) = _____

i) 0.07746 (3dp) = _____

j) 0.895 (2dp) = _____

k) 7.9601 (1dp) = _____

l) 0.6703 (3dp) = _____

2. Write a possible question for each answer

a) _____ (2dp) = 5.82

b) _____ (1dp) = 3.5

c) _____ (3dp) = 0.094

d) _____ (1dp) = 45.0

e) _____ (2dp) = 0.70

Rounding to decimal places

1. Round each number to the required number of decimal places (dp)

a) 7.842 (1dp) = **7.8**

b) 45.65 (1dp) = **45.7**

c) 0.109 (1dp) = **0.1**

d) 4.817 (2dp) = **4.82**

e) 12.969 (2dp) = **12.97**

f) 0.0814 (2dp) = **0.08**

g) 5.49284 (3dp) = **5.493**

h) 0.007891 (3dp) = **0.008**

i) 0.07746 (3dp) = **0.077**

j) 0.895 (2dp) = **0.90**

k) 7.9601 (1dp) = **8.0**

l) 0.6703 (3dp) = **0.670**

2. Write a possible question for each answer

a) _____ (2dp) = 5.82 **5.815 to 5.825**

b) _____ (1dp) = 3.5 **3.45 to 3.55**

c) _____ (3dp) = 0.094 **0.0935 to 0.0945**

d) _____ (1dp) = 45.0 **44.95 to 45.05**

e) _____ (2dp) = 0.70 **0.695 to 0.705**

Rounding to significant figures

If your salary is £37,281 which is the most significant figure?

The 3 represents £30,000

The 7 represents £7,000

The 2 represents £200

The 8 represents £80

The 1 represents £1

Clearly the 3 is the most significant, as it is the digit on the left, and so has the highest **place value**

The first non-zero digit is called the **1st significant figure**

Digits after this, including zeros, are then called the 2nd, 3rd, 4th, … significant figures

1st significant figure
2nd significant figure
3rd significant figure

Eg 6 7 0 4

1st significant figure
2nd significant figure
3rd significant figure

Eg 0 . 0 2 9 5

When rounding to significant figures, use the same system as used for rounding to decimal places

Rounding to significant figures

Eg round 7.496 to 1 significant figure

1st significant figure | 2nd significant figure | 3rd significant figure

7 . 4 9 6 = 7 (1sf)

Don't change

5 or more?

Imagine line after desired significant figure

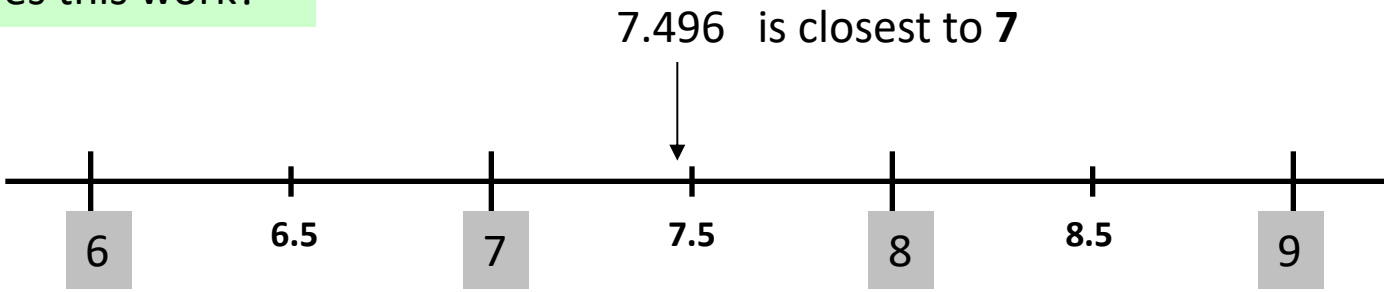
If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

Write a rounded value with zeros filled in for any digits after the line but before the decimal point

In this question, you effectively need to round 7.496 to the nearest **whole number**:

Why does this work?



Rounding to significant figures

Eg round 3857 to 2 significant figures

1st significant figure | 2nd significant figure | 3rd significant figure

3 8 5 7 = 3900 (2sf)

Increase by 1

5 or more?

Imagine line after desired significant figure

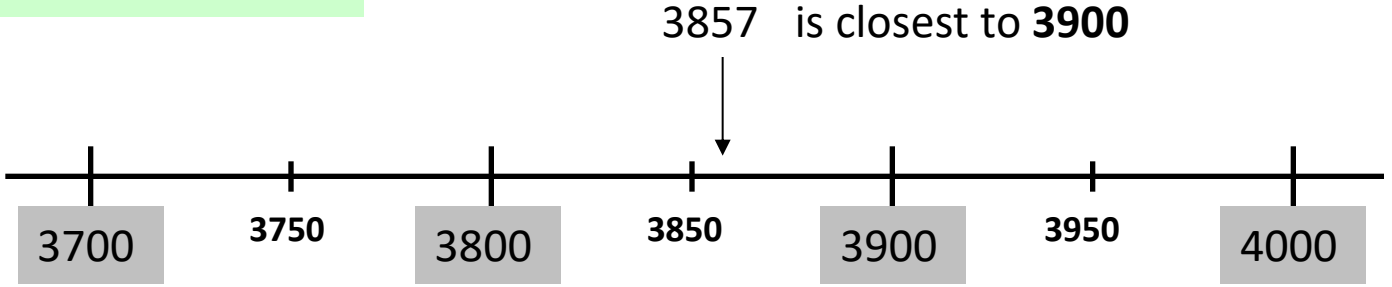
If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

Write a rounded value with zeros filled in for any digits after the line but before the decimal point

In this question, you effectively need to round 3857 to the nearest **hundred**:

Why does this work?



Rounding to significant figures

Eg round 0.0679 to 1 significant figure

1st significant figure | 2nd significant figure | 3rd significant figure

The first *non-zero* digit is the 1st significant figure

0 . 0 6 7 9 = 0.07 (1sf)

Increase by 1

5 or more?

Imagine line after desired significant figure

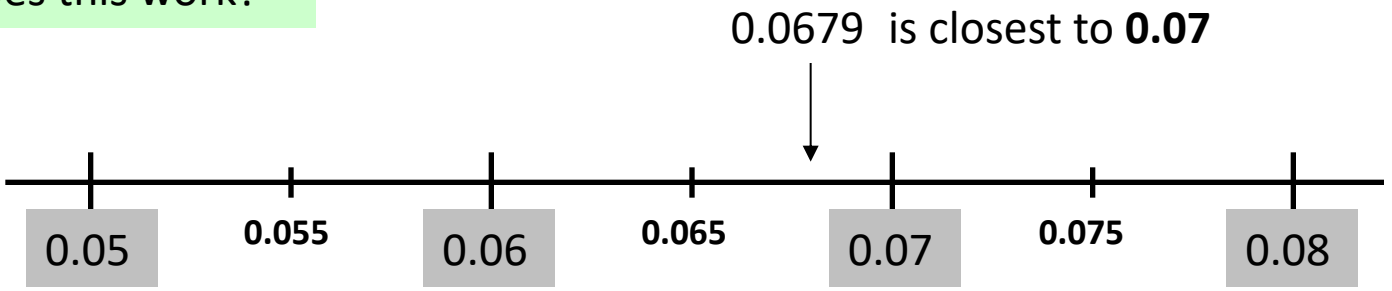
If the number after the line is 5 or more, increase the value before the line by 1

If not, don't change the value before the line

Write a rounded value with zeros filled in for any digits after the line but before the decimal point

In this question, you effectively need to round 0.0679 to the nearest **hundredth:**

Why does this work?





Rounding to significant figures



1. Round each number to the required number of significant figures (sf)

a) 7.842 (1sf) = _____

b) 45.65 (1sf) = _____

c) 628 (1sf) = _____

d) 4887 (2sf) = _____

e) 12.969 (2sf) = _____

f) 0.0814 (2sf) = _____

g) 5.49284 (3sf) = _____

h) 0.007891 (1sf) = _____

i) 0.07746 (3sf) = _____

j) 0.895 (2sf) = _____

k) 7.9601 (1sf) = _____

l) 0.6703 (3sf) = _____

2. Write a possible question for each answer

a) _____ (2sf) = 580

b) _____ (1sf) = 3000

c) _____ (3sf) = 12.5

d) _____ (1sf) = 0.09

e) _____ (2sf) = 6.0

Rounding to significant figures

1. Round each number to the required number of significant figures (sf)

a) 7.842 (1sf) = **8**

b) 45.65 (1sf) = **50**

c) 628 (1sf) = **600**

d) 4887 (2sf) = **4900**

e) 12.969 (2sf) = **13**

f) 0.0814 (2sf) = **0.081**

g) 5.49284 (3sf) = **5.49**

h) 0.007891 (1sf) = **0.008**

i) 0.07746 (3sf) = **0.0775**

j) 0.895 (2sf) = **0.90**

k) 7.9601 (1sf) = **8**

l) 0.6703 (3sf) = **0.670**

2. Write a possible question for each answer

a) _____ (2sf) = 580 **575 to 585**

b) _____ (1sf) = 3000 **2500 to 3500**

c) _____ (3sf) = 12.5 **12.45 to 12.55**

d) _____ (1sf) = 0.09 **0.085 to 0.095**

e) _____ (2sf) = 6.0 **5.95 to 6.05**

Rounding to significant figures (sf)

1st sf	2nd sf	3rd sf	4th sf



Estimation



The method of rounding to significant figures can be used to estimate calculations

Round each number in the calculation to 1 significant figure

Work out the calculation

Remember to follow BIDMAS

You may need to divide by a number less than 1

It may help to use an equivalent fraction

Eg estimate $\dfrac{2371\times 1.84}{36.92} \approx \dfrac{2000\times 2}{40} = \dfrac{4000}{40} = 100$

Eg estimate $\dfrac{4.81+3.31^2}{17.29+8.43} \approx \dfrac{5+3^2}{20+8} = \dfrac{14}{28} = 0.5$

Eg estimate $\dfrac{28.8-17.36}{0.184} \approx \dfrac{30-20}{0.2} = \dfrac{10}{0.2} = \dfrac{100}{2} = 50$

Eg estimate $\dfrac{6.69-1.28}{0.315} \approx \dfrac{7-1}{0.3} = \dfrac{6}{0.3} = \dfrac{60}{3} = 20$



Estimation



1. Work out an estimate for the value of:

a) $\dfrac{637}{3.2\times 9.8}$

b) $\dfrac{29.3\times 10.3^2}{48.76}$

c) $\dfrac{7813\div 376}{4.68\times 8.4}$

d) $\dfrac{5.79\times 312}{0.523}$

e) $\dfrac{421\times 5.904}{0.195}$

f) $\dfrac{23.7\times 2.85}{0.618}$

2. On average, Nick walks 18 000 steps every day.
He walks 1 mile approximately every 3500 steps.

Work out an estimate for the average distance, in miles, that Nick walks **in one year.**



Estimation



1. Work out an estimate for the value of:

a) $\frac{637}{3.2\times 9.8} \approx \frac{600}{3\times 10} = \frac{600}{30} = 20$

b) $\frac{29.3\times 10.3^2}{48.76} \approx \frac{30\times 100}{50} = 30\times 2 = 60$

c) $\frac{7813\div 376}{4.68\times 8.4} \approx \frac{8000\div 400}{5\times 8} = \frac{20}{40} = \frac{1}{2}$

d) $\frac{5.79\times 312}{0.523} \approx \frac{6\times 300}{0.5} = 1800\times 2 = 3600$

e) $\frac{421\times 5.904}{0.195} \approx \frac{400\times 6}{0.2} = \frac{2400}{0.2} = \frac{24000}{2} = 12000$

f) $\frac{23.7\times 2.85}{0.618} \approx \frac{20\times 3}{0.6} = \frac{60}{0.6} = \frac{600}{6} = 100$

2. On average, Nick walks 18 000 steps every day.
He walks 1 mile approximately every 3500 steps.

Work out an estimate for the average distance, in miles, that Nick walks **in one year.**

$$\frac{18000\times 365}{3500} \approx \frac{20000\times 400}{4000} = 2000$$

3. Look at the following questions and answers. Some of the answers are wrong. Decide which ones are probably wrong by making an estimate.

Question	Answer given	Estimate	Sensible answer given?
245 x 19	48365		
12% of 192	23.04		
5.12 x 38.76	124.42		
29% of 305	47.84		
0.061×43	2.623		
4829×0.00715	34.5273		

Estimation

3. Look at the following questions and answers. Some of the answers are wrong. Decide which ones are probably wrong by making an estimate.

Question	Answer given	Estimate	Sensible answer given?
245 x 19	48365	$200 \times 20 = 4000$	No
12% of 192	23.04	10% of 200 $= 20$	Yes
5.12 x 38.76	124.42	$5 \times 40 = 200$	No
29% of 305	47.84	30% of 300 $= 90$	No
0.061×43	2.623	$0.06 \times 40 = 0.6 \times 4 = 2.4$	Yes
4829×0.00715	34.5273	$5000 \times 0.007 = 5 \times 7 = 35$	Yes

$9 \times 5 = 45$ The answer **must** end in the digit 5

Shock keeper

Some unscrupulous shop keepers take advantage of our assumption that the till will always give the correct change. Others advertise offers that, when you actually go to pay, don't get honoured.

Being able to **estimate** your bill and the correct change can not only help budgeting but ensure these 'mistakes' don't occur!

Eg I go to two shops and buy the following:

Shop A

Pizza £3.49
Salad £1.69
Chocolate cake £2.25

I pay with £10 and get about £2.50 change

Shop B

Washing powder £6.99
Shampoo £5.49
Conditioner £4.35

I pay with £20 and get about £2 change

Which shop has short-changed me?!

highest and lowest bounds



newspapers regularly approximate numbers
what is the smallest and largest you think the actual numbers could have been?

(1) a man received £500 for unfair dismissal

(2) attackers stole £35 in cash

(3) the air ambulance took 20 minutes

(4) someone survived a 400 ft. fall into snow at Morte Base

(5) "I knew Jean Kent for 50 years"

(6) a salary of £24K

(7) the collision happened around 2.30 a.m.

(8) the Jaguar factory will create 700 jobs

(9) 33 held in a 1000 person EU protest in the Ukraine

(10) rowers embark on 3000 mile Atlantic charity bid

(11) appeal has raised £34,500

(12) unemployment is currently around 2.5 million

(13) average daily sales of three quarters of a million

(14) a £1.7M visitor centre scheme

(15) Margaret Thatcher had a £12 million house in a tax haven

(16) the US Government agrees to settle a $50 million lawsuit

(17) in 1982 New Zealand had 70 million sheep and 3.2 million people

(18) the estimated 25,000 polar bear population are a species of special concern

(19) £200,000 car vandalism spree

(20) 25,000 waiting list for social housing

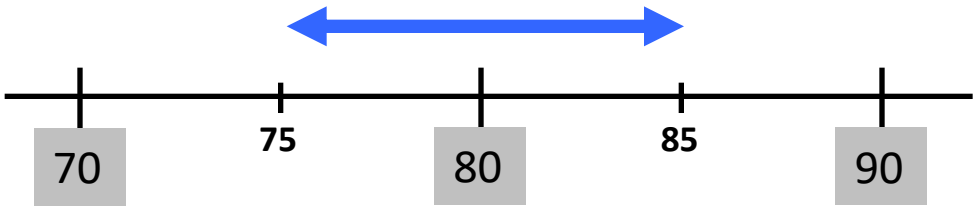
Upper and lower bounds

Bob weighs himself as 80kg, correct to the nearest ten kg
What is the least he could actually weigh? And the most?

The least he could weigh is 75kg

He could weigh anything between 75 and 85kg

The most he could weigh is 85kg

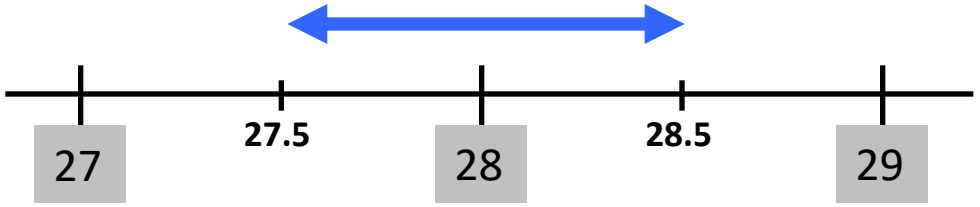


Dave draws an angle of 28°, correct to the nearest degree.
What is the smallest the angle could actually be? And the biggest?

The smallest is 27.5°

The actual angle drawn could be anything between 27.5 and 28.5°

The biggest is 28.5°



Upper and lower bounds

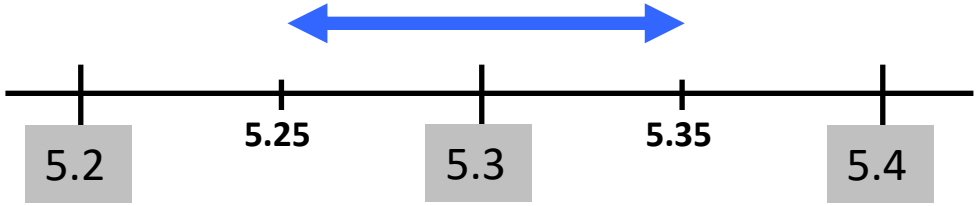
The smallest value an approximation could actually be is called the **lower bound**

The largest value an approximation could actually be is called the **upper bound**

Eg a number is rounded to 5.3 (1 decimal place)
What are the upper and lower bounds?

Lower bound = 5.25
Upper bound = 5.35

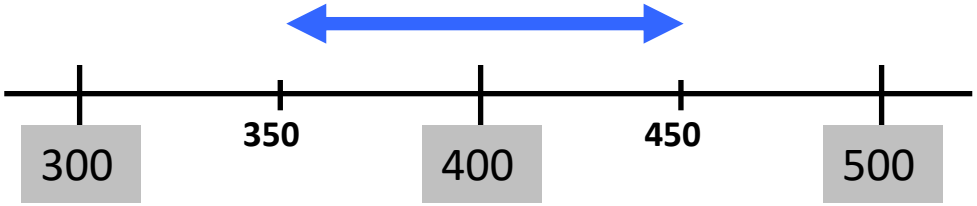
1dp = nearest 10th



Eg a number is rounded to 400 (1 significant figure)
What are the upper and lower bounds?

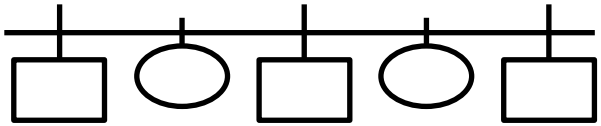
Lower bound = 350
Upper bound = 450

1sf = nearest 100



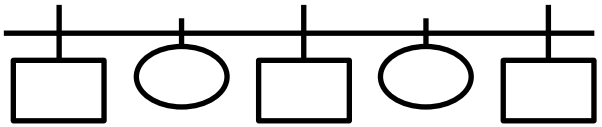
Upper and lower bounds

a) 80 (nearest ten)



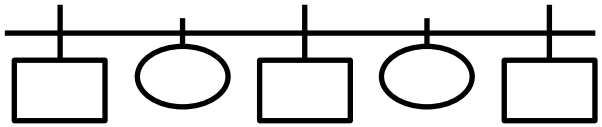
Lower bound = Upper bound =

b) 2300 (nearest hundred)



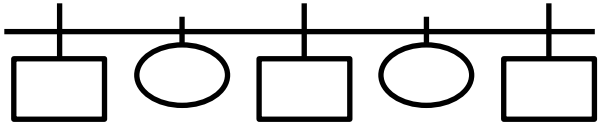
Lower bound = Upper bound =

c) 35 (nearest five)



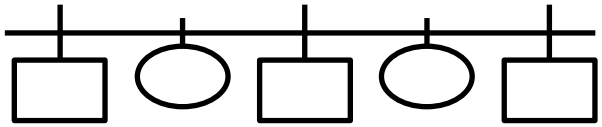
Lower bound = Upper bound =

d) 4 (nearest whole number)



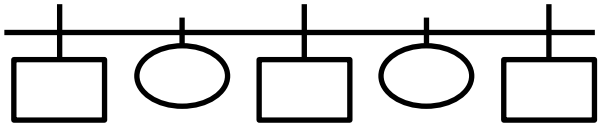
Lower bound = Upper bound =

e) 700 (nearest ten)



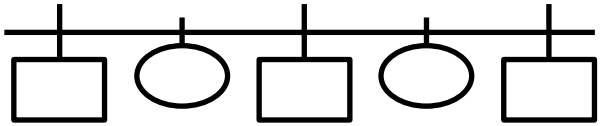
Lower bound = Upper bound =

f) 2000 (nearest hundred)



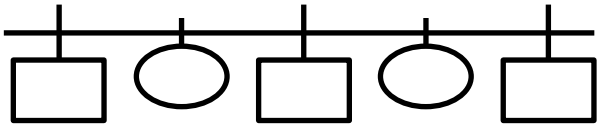
Lower bound = Upper bound =

g) 4.7 (nearest tenth)



Lower bound = Upper bound =

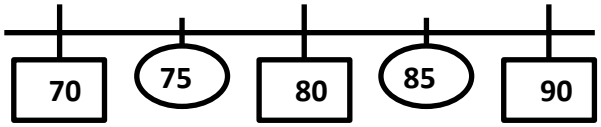
h) 0.36 (nearest hundredth)



Lower bound = Upper bound =

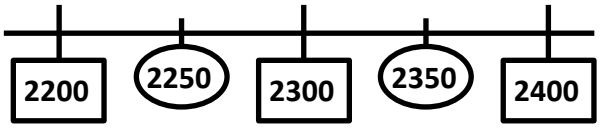
Upper and lower bounds

a) 80 (nearest ten)



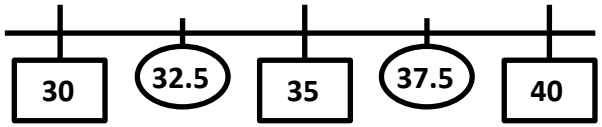
Lower bound = **75** Upper bound = **85**

b) 2300 (nearest hundred)



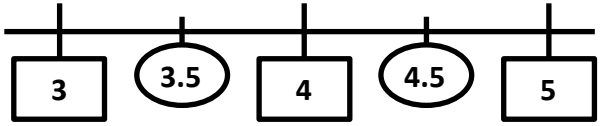
Lower bound = **2250** Upper bound = **2350**

c) 35 (nearest five)



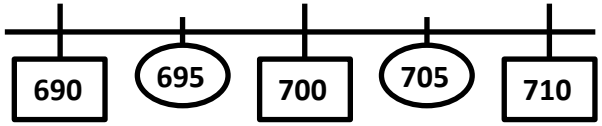
Lower bound = **32.5** Upper bound = **37.5**

d) 4 (nearest whole number)



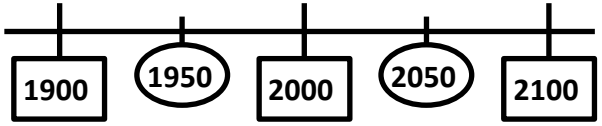
Lower bound = **3.5** Upper bound = **4.5**

e) 700 (nearest ten)



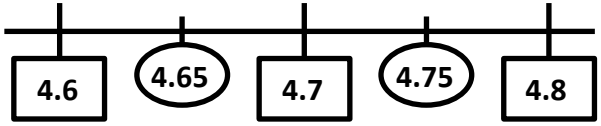
Lower bound = **695** Upper bound = **705**

f) 2000 (nearest hundred)



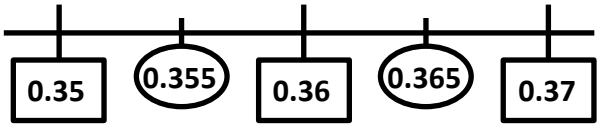
Lower bound = **1950** Upper bound = **2050**

g) 4.7 (nearest tenth)

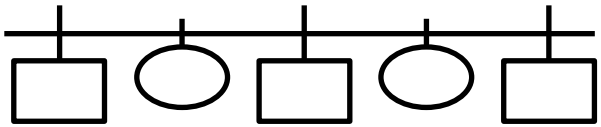
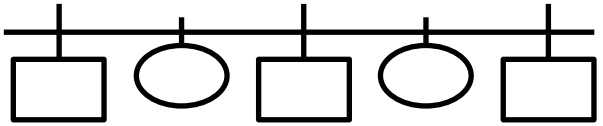
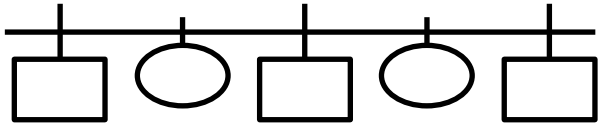
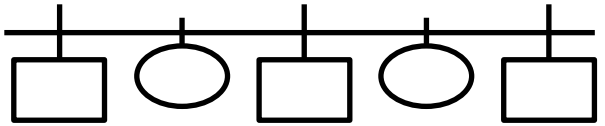
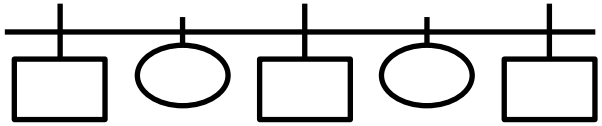
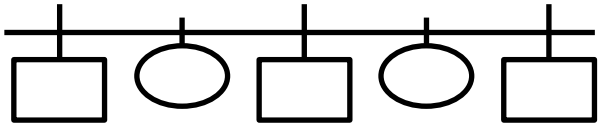
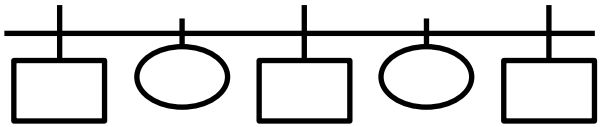
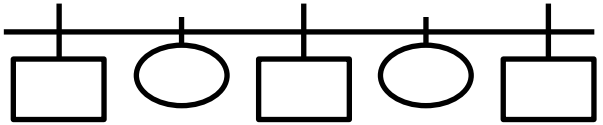


Lower bound = **4.65** Upper bound = **4.75**

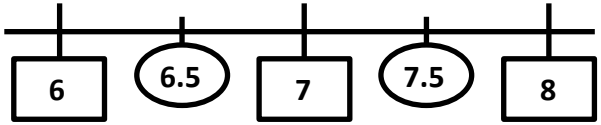
h) 0.36 (nearest hundredth)



Lower bound = **0.355** Upper bound = **0.365**

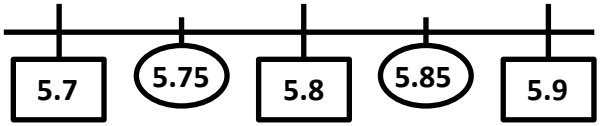
i) 7 (1 significant figure)  Lower bound = Upper bound =	j) 5.8 (1 decimal place)  Lower bound = Upper bound =
k) 30 (1significant figure)  Lower bound = Upper bound =	l) 610 (2 significant figures)  Lower bound = Upper bound =
m) 0.04 (1 significant figure)  Lower bound = Upper bound =	n) 0.84 (2 decimal places)  Lower bound = Upper bound =
o) 5.0 (1 decimal place)  Lower bound = Upper bound =	p) 70 (2 significant figures)  Lower bound = Upper bound =

i) 7 (1 significant figure)



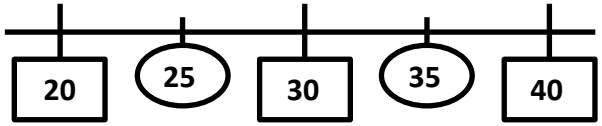
Lower bound = **6.5** Upper bound = **7.5**

j) 5.8 (1 decimal place)



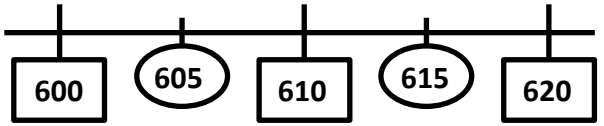
Lower bound = **5.75** Upper bound = **5.85**

k) 30 (1significant figure)



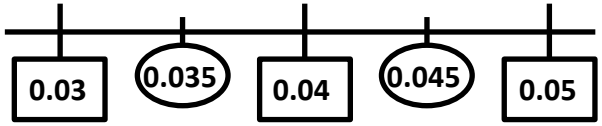
Lower bound = **25** Upper bound = **35**

l) 610 (2 significant figures)



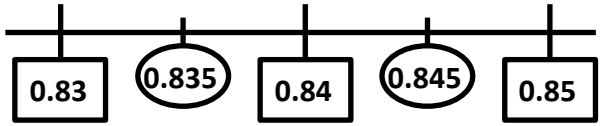
Lower bound = **605** Upper bound = **615**

m) 0.04 (1 significant figure)



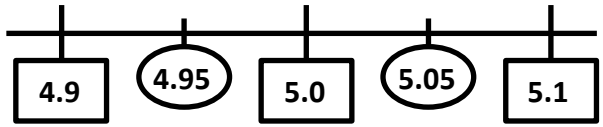
Lower bound = **0.035** Upper bound = **0.045**

n) 0.84 (2 decimal places)



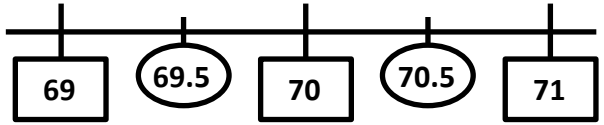
Lower bound = **0.835** Upper bound = **0.845**

o) 5.0 (1 decimal place)



Lower bound = **4.95** Upper bound = **5.05**

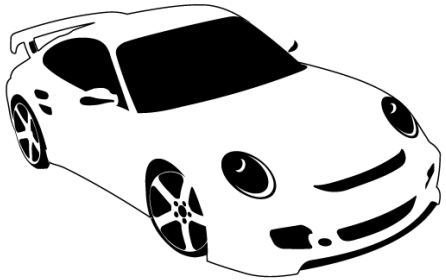
p) 70 (2 significant figures)



Lower bound = **69.5** Upper bound = **70.5**

Upper and lower bounds

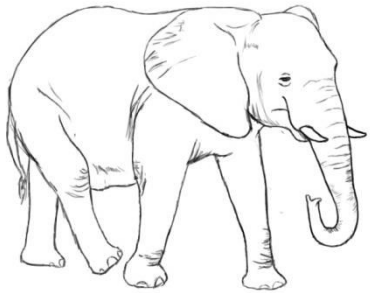
upper bound =



Top speed = 140mph
(nearest 10)

lower bound =

upper bound =



Weight of African elephant
= 2700kg (2 significant figures)

lower bound =

upper bound =



Length of green Anaconda
= 5m (1 significant figure)

lower bound =

upper bound =



Average UK salary
= £26500 (nearest 500)

lower bound =

upper bound =



Age of Earth = 4.5 billion years
(2 significant figures)

lower bound =

upper bound =

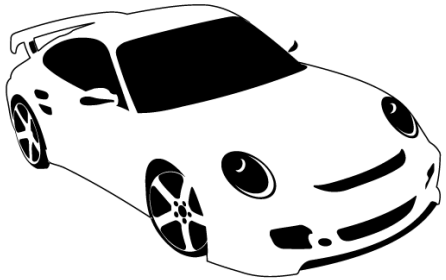


Volume of blood in human body =
4.7 litres (1 decimal place)

lower bound =

Upper and lower bounds

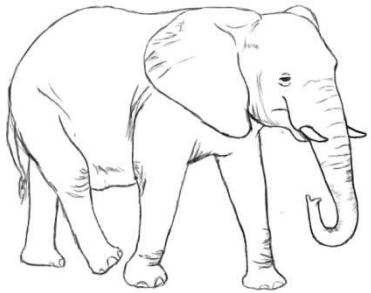
upper bound = **145**



Top speed = 140mph
(nearest 10)

lower bound = **135**

upper bound = **2750**



Weight of African elephant
= 2700kg (2 significant figures)

lower bound = **2650**

upper bound = **5.5**



Length of green Anaconda
= 5m (1 significant figure)

lower bound = **4.5**

upper bound = **26750**



Average UK salary
= £26500 (nearest 500)

lower bound = **26250**

upper bound = **4.55**



Age of Earth = 4.5 billion years
(2 significant figures)

lower bound = **4.45**

upper bound = **4.75**



Volume of blood in human body =
4.7 litres (1 decimal place)

lower bound = **4.65**

A rectangle is measured as 4cm by 3cm, both measurements correct to 1 significant figure.



What sums would you do to find the upper and lower bounds of the perimeter and the area of the rectangle?

	Lower bound	Upper bound
Perimeter	$2\times(2.5+3.5)$	$2\times(3.5+4.5)$
Area	2.5×3.5	3.5×4.5

Complete the table

calculation	Lower bound	Upper bound
$a+b$	$a_{\min}+b_{\min}$	$a_{\max}+b_{\max}$
$a-b$	$a_{\min}-b_{\max}$	$a_{\max}-b_{\min}$
ab	$a_{\min}\times b_{\min}$	$a_{\max}\times b_{\max}$
$\frac{a}{b}$	$\frac{a_{\min}}{b_{\max}}$	$\frac{a_{\max}}{b_{\min}}$

$$x = a + b$$

Write down the sum needed to find the **smallest** possible value of x if:

a = 6 (1sf)

b = 2 (1sf)

$5.5 + 1.5$

$$x = a + b$$

Write down the sum needed to find the **largest** possible value of x if:

a = 6 (1sf)

b = 2 (1sf)

$$6.5 + 2.5$$

$$x = a - b$$

Write down the sum needed to find the **largest** possible value of x if:

a = 3.7 (1dp)

b = 0.8 (1dp)

$3.75 - 0.75$

$$x = a - b$$

Write down the sum needed to find the **smallest** possible value of x if:

a = 3.7 (1dp)

b = 0.8 (1dp)

$3.65 - 0.85$

$$x = ab$$

Write down the sum needed to find the **largest** possible value of x if:

a = 50 (2sf)

b = 14 (2sf)

50.5×14.5

$$x = ab$$

Write down the sum needed to find the **smallest** possible value of x if:

a = 50 (2sf)

b = 14 (2sf)

49.5×13.5

$$x = \frac{a}{b}$$

Write down the sum needed to find the **smallest** possible value of x if:

a = 15.0 (1dp)

b = 2.82 (2dp)

$$\frac{14.95}{2.825}$$

$$x = \frac{a}{b}$$

Write down the sum needed to find the **largest** possible value of x if:

a = 15.0 (1dp)

b = 2.82 (2dp)

$$\frac{15.05}{2.815}$$

Upper and lower bounds of calculations

Complete the table. The first question has been done for you as an example.

a	Bounds of a	b	Bounds of b	calculation	Lower bound of calculation	Upper bound of calculation
8 (1sf)	Upper = 8.5	3 (1sf)	Upper = 3.5	$a+b$	$a_{\min}+b_{\min}$ $=7.5+2.5$	$a_{\max}+b_{\max}$ $=8.5+3.5$
	Lower = 7.5		Lower = 2.5			
4.2 (1dp)	Upper =	3.8 (1dp)	Upper =	$a-b$		
	Lower =		Lower =			
35 (2sf)	Upper =	1.7 (2sf)	Upper =	ab		
	Lower =		Lower =			
0.6 (1sf)	Upper =	0.09 (1sf)	Upper =	$\frac{a}{b}$		
	Lower =		Lower =			

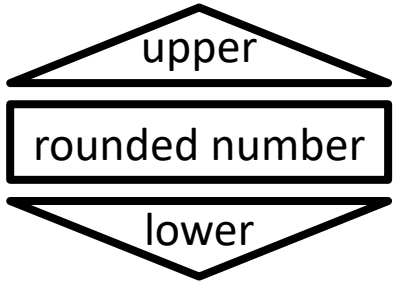
Upper and lower bounds of calculations

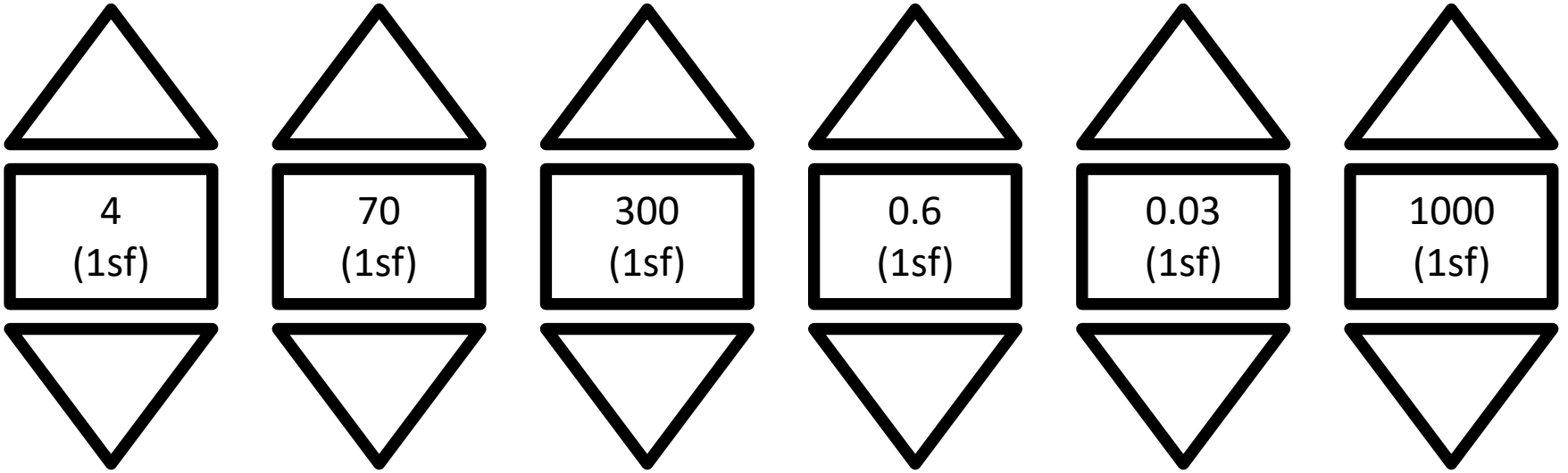
Complete the table. The first question has been done for you as an example.

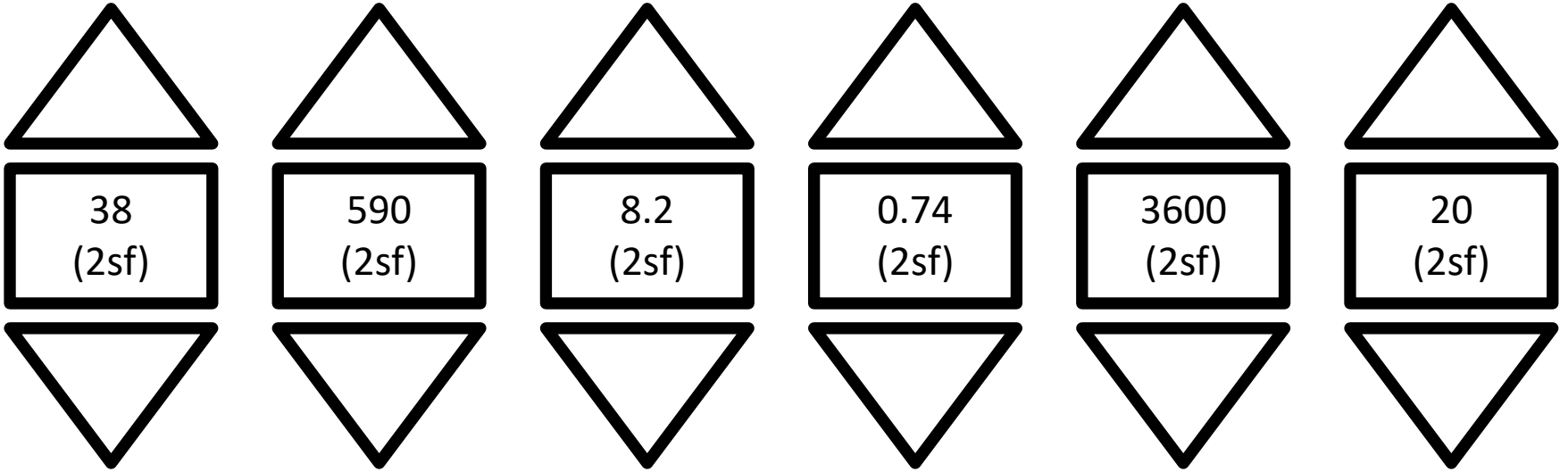
a	Bounds of a	b	Bounds of b	calculation	Lower bound of calculation	Upper bound of calculation
8 (1sf)	Upper = 8.5	3 (1sf)	Upper = 3.5	$a+b$	$a_{\min}+b_{\min}$ $=7.5+2.5$	$a_{\max}+b_{\max}$ $=8.5+3.5$
	Lower = 7.5		Lower = 2.5			
4.2 (1dp)	Upper = 4.25	3.8 (1dp)	Upper = 3.85	$a-b$	$a_{\min}-b_{\max}$ $=4.15-3.85$	$a_{\max}-b_{\min}$ $=4.25-3.75$
	Lower = 4.15		Lower = 3.75			
35 (2sf)	Upper = 35.5	1.7 (2sf)	Upper = 1.75	ab	$a_{\min}\times b_{\min}$ $=34.5\times 1.65$	$a_{\max}\times b_{\max}$ $=35.5\times 1.75$
	Lower = 34.5		Lower = 1.65			
0.6 (1sf)	Upper = 0.65	0.09 (1sf)	Upper = 0.095	$\frac{a}{b}$	$\frac{a_{\min}}{b_{\max}}=\frac{0.55}{0.095}$	$\frac{a_{\max}}{b_{\min}}=\frac{0.65}{0.085}$
	Lower = 0.55		Lower = 0.085			

Upper and lower bounds

Write down the upper and lower bound to each rounded number

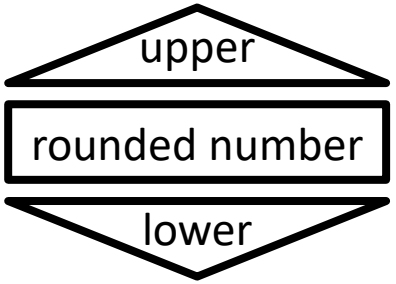






Upper and lower bounds

Write down the upper and lower bound to each rounded number



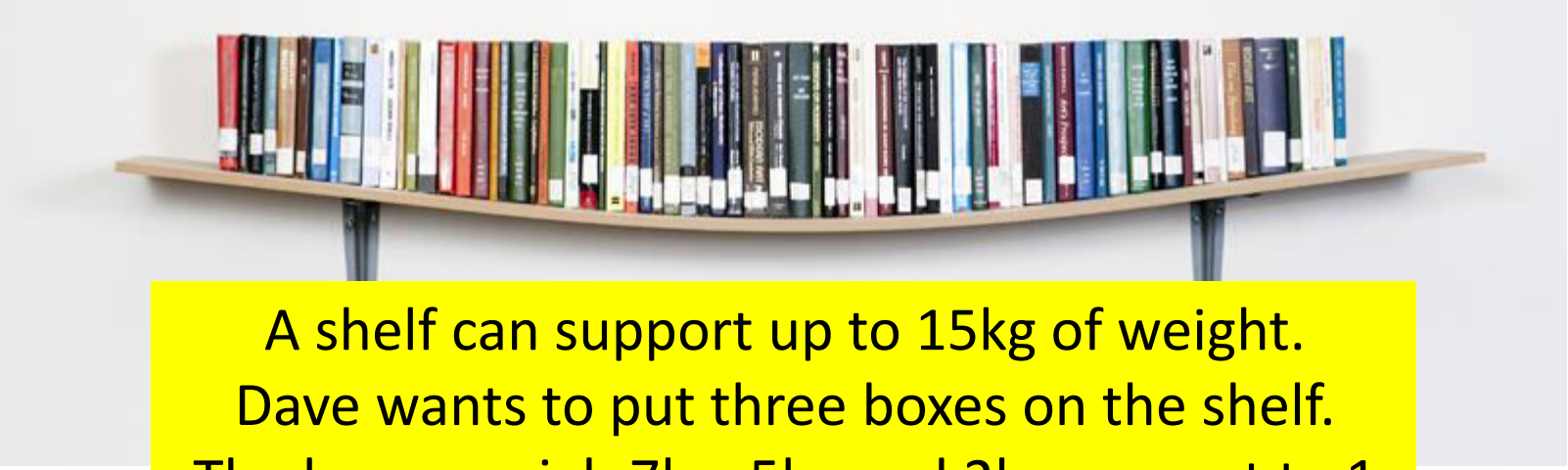
4.5	75	350	0.65	0.035	1500
4 (1sf)	70 (1sf)	300 (1sf)	0.6 (1sf)	0.03 (1sf)	1000 (1sf)
3.5	65	250	0.55	0.025	500
38.5	595	8.25	0.745	3650	20.5
38 (2sf)	590 (2sf)	8.2 (2sf)	0.74 (2sf)	3600 (2sf)	20 (2sf)
37.5	585	8.15	0.735	3550	19.5

Error intervals

Use your answers to the last worksheet to fill in the table

Rounded value	Lower bound	Upper bound	Error interval
4 (1sf)	3.5	4.5	$3.5 \leq$ actual value < 4.5

Bounds in context



A shelf can support up to 15kg of weight.
Dave wants to put three boxes on the shelf.
The boxes weigh 7kg, 5kg and 2kg, correct to 1 significant figure.
Is there any chance the shelf could break when Dave puts the boxes on it?

Upper bound of weight of boxes = 7.5+5.5+2.5 = 15.5kg

The shelf could break!



Bounds in context



$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Mr Walker is going to run the marathon.
To achieve his target time, he needs to average 10 km/h

In training, he runs 15 kilometres (to the nearest kilometre) in 90 minutes (to the nearest 5 minutes).

Is this definitely in line with his target?!

$$\text{speed}_{\text{min}} = \frac{\text{distance}_{\text{min}}}{\text{time}_{\text{max}}}$$

$$= \frac{14.5}{1.541\ldots} = 9.405\ldots \text{km/h}$$

$$\text{distance}_{\text{min}} = 14.5\text{km}$$

$$\text{time}_{\text{max}} = \frac{92.5}{60} = 1.541\ldots \text{hours}$$

So not in line with the target!

Bounds in context

1. Peter transports metal bars in his van.
The van has a safety notice “Maximum Load 1200 kg”.
Each metal bar has a label “Weight 60 kg”.
For safety reasons Peter assumes that 1200 is rounded correct to 2 significant figures
and 60 is rounded correct to 1 significant figure.
Calculate the greatest number of bars that Peter can **safely** put into the van if his assumptions are correct.

Bounds in context

1. Peter transports metal bars in his van.
The van has a safety notice “Maximum Load 1200 kg”.
Each metal bar has a label “Weight 60 kg”.
For safety reasons Peter assumes that 1200 is rounded correct to 2 significant figures and 60 is rounded correct to 1 significant figure.
Calculate the greatest number of bars that Peter can **safely** put into the van if his assumptions are correct.

$$\text{Number of bars} = \frac{\text{Van capacity}}{\text{Weight of each bar}}$$

Worst case scenario is that van capacity is lower than stated and bars are heavier than stated

$$\text{Lower bound} = \frac{\text{Van capacity}_{\text{min}}}{\text{Weight of each bar}_{\text{max}}} = \frac{1150}{65} = 17.69...$$

So the greatest number of bars that Peter can **safely** put into the van is **17**

2. An aeroplane travels at 560mph, correct to 2 significant figures.
It travels exactly 8500 miles from Dallas to Sydney.
Work out the difference between the longest and shortest time it could take to do this journey, correct to the nearest minute.

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It travels exactly 8500 miles from Dallas to Sydney.
Work out the difference between the longest and shortest time it could take to do this journey, correct to the nearest minute.

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\text{time}_{\text{max}} = \frac{\text{distance}}{\text{speed}_{\text{min}}} = \frac{8500}{555}$$

$$\text{time}_{\text{min}} = \frac{\text{distance}}{\text{speed}_{\text{max}}} = \frac{8500}{565}$$

$$\text{difference} = \frac{8500}{555} - \frac{8500}{565} = 0.271\ldots \text{ hours } = 0.271\ldots \times 60 \approx 16 \text{ minutes}$$

3. A company manufactures bricks.
The bricks are cuboids in shape, measuring 225mm long by 112mm wide by 75mm high.
The machine which makes them is accurate to the nearest mm.

a) Work out the biggest and smallest possible volumes of a brick, giving your answer correct to the nearest mm^3.

b) The bricks are used to build a wall 50 metres long and 3 metres high.
Work out the largest number of bricks that could be needed.

3. A company manufactures bricks.
The bricks are cuboids in shape, measuring 225mm long by 112mm wide by 75mm high.
The machine which makes them is accurate to the nearest mm.
a) Work out the biggest and smallest possible volumes of a brick, giving your answer correct to the nearest mm^3.

$$\text{volume}_{\text{max}} = 225.5 \times 112.5 \times 75.5 = 1915341\text{mm}^3$$

$$\text{volume}_{\text{min}} = 224.5 \times 111.5 \times 74.5 = 1864865\text{mm}^3$$

b) The bricks are used to build a wall 50 metres long and 3 metres high.
Work out the largest number of bricks that could be needed.

$50000 \div 224.5 = 222.71...$ so 223 bricks long

$3000 \div 74.5 = 40.26...$ so 41 bricks high

$$223 \times 41 = 9143$$

4. A car travels a distance of 34.5 miles to the nearest 0.1 mile using 1.4 gallons of petrol to the nearest 0.1 gallon.

$$\text{fuel economy} = \frac{\text{distance travelled}}{\text{petrol used}}$$

What is:

(i) the highest possible rate of fuel economy and

(ii) the lowest possible rate ?

Give your answers correct to 1 decimal place.

4. A car travels a distance of 34.5 miles to the nearest 0.1 mile using 1.4 gallons of petrol to the nearest 0.1 gallon.

$$\text{fuel economy} = \frac{\text{distance travelled}}{\text{petrol used}}$$

What is:
(i) the highest possible rate of fuel economy and
(ii) the lowest possible rate ?
Give your answers correct to 1 decimal place.

$$\text{fuel economy}_{\text{max}} = \frac{\text{distance travelled}_{\text{max}}}{\text{petrol used}_{\text{min}}} = \frac{34.55}{1.35} = 25.6 \text{ miles per gallon (1dp)}$$

$$\text{fuel economy}_{\text{min}} = \frac{\text{distance travelled}_{\text{min}}}{\text{petrol used}_{\text{max}}} = \frac{34.45}{1.45} = 23.8 \text{ miles per gallon (1dp)}$$